

# Combinatorial Geometry

Labix

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# 1 Polyhedrons and Polytopes

## 1.1 Polyhedrons and its Properties

Polytopes exists more generically in Euclidean space. We lose nothing discussing polytopes in  $\mathbb{R}^n$  since Euclidean spaces maybe identified with  $\mathbb{R}^n$  canonically.

### Definition 1.1.1 (Halfspaces)

A halfspace of  $\mathbb{R}^n$  is a subset of  $\mathbb{R}^n$  of the form

$$H_{x,b}^+ = \{x \in \mathbb{R}^n \mid \langle a, x \rangle \geq -b\}$$

for some  $a, b \in \mathbb{R}^n$ .

### Definition 1.1.2 (Polyhedrons)

Let  $P \subseteq \mathbb{R}^n$ . We say that  $P$  is a polyhedron if there exists finitely many half spaces  $H_{x_1,b_1}^+, \dots, H_{x_k,b_k}^+$  such that

$$P = \bigcap_{i=1}^k H_{x_i,b_i}^+$$

### Definition 1.1.3 (Dimension)

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. Define the dimension of  $P$  to be the dimension of the smallest affine subspace containing  $P$ .

### Definition 1.1.4 (Full-Dimensional Polyhedrons)

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. We say that  $P$  is full-dimensional if  $\dim(P) = n$ .

### Definition 1.1.5 (Polytopes)

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. We say that  $P$  is a polytope if  $P$  is bounded.

### Theorem 1.1.6 (Weyl-Minkowski)

Let  $P \subseteq \mathbb{R}^n$  be a subset. Then  $P$  is a polytope if and only if there exists a finite subset  $S \subseteq \mathbb{R}^n$  such that  $P = \text{Conv}(S)$ .

## 1.2 The Faces of a Polyhedron

### Definition 1.2.1 (Supporting Hyperplane)

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. Let  $H \subseteq \mathbb{R}^n$  be a hyperplane. We say that  $H$  is a supporting hyperplane for  $P$  if the following are true.

- $P \cap H \neq \emptyset$ .
- $P$  lies entirely in one of the halfspaces defined by  $H$ .

### Definition 1.2.2 (Faces)

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. A face of  $P$  is a subset of  $P$  defined by

$$F = P \cap H$$

where  $H$  is a supporting hyperplane for  $P$ .

### Lemma 1.2.3

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. Let  $F \subseteq P$  be a face. Then  $F$  is a polytope.

### Lemma 1.2.4

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. Let  $F, G \subseteq P$  be faces of  $P$ . Then  $F \cap G$  is a face of  $P$ .

**Lemma 1.2.5**

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. Let  $F \subseteq P$  be a face of  $P$ . Let  $G \subseteq F$  be a face of  $F$ . Then  $G$  is a face of  $P$ .

**Definition 1.2.6 (Vertices, Edges and Facets)**

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. Let  $F \subseteq P$  be a face.

- We say that  $F$  is a vertex if  $\dim(F) = 0$ . Denote the set of vertices of  $P$  by  $V(P)$ .
- We say that  $F$  is an edge if  $\dim(F) = 1$ .
- We say that  $F$  is a facet if  $\dim(F) = \dim(P) - 1$ .

**Definition 1.2.7 (The Face Poset)**

Let  $P \subseteq \mathbb{R}^n$  be a polyhedron. Define the face poset  $\mathcal{F}(P)$  of  $P$  to be the set of faces of  $P$  together with inclusion.

**Definition 1.2.8 (The Face Vector)**

Let  $P \subseteq \mathbb{R}^n$  be a polytope of dimension  $d$ . Define the face vector  $(f_0, \dots, f_{d-1})$  of  $S$  by the formula

$$f_i = \text{The number of } i\text{-dimensional faces of } P$$

for  $0 \leq i \leq d-1$  and by convention,  $f_{-1} = f_d = 1$ .

**Lemma 1.2.9 (Euler's Formula)**

Let  $P \subseteq \mathbb{R}^n$  be a polytope of dimension  $d$ . Then we have

$$\sum_{i=0}^{d-1} (-1)^i f_i = 1 + (-1)^d$$

**Definition 1.2.10 (Combinatorially Equivalent)**

Let  $P_1 \subseteq \mathbb{R}^{n_1}$  and  $P_2 \subseteq \mathbb{R}^{n_2}$  be polytopes. We say that  $P_1$  and  $P_2$  are combinatorially equivalent if there exists a inclusion preserving bijection between  $\mathcal{F}(P_1)$  and  $\mathcal{F}(P_2)$ . In this case we write  $P_1 \simeq P_2$ .

**Definition 1.2.11 (Edge Graph)**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Define the edge graph  $G(P)$  of  $P$  by the following data.

- $V(G(P)) = V(P)$ .
- $E(G(P)) = \{(p, q) \mid \text{There exists an edge } E \text{ such that } \partial E = \{p, q\}\}$ .

### 1.3 Properties of Convex Polytopes

**Proposition 1.3.1**

Let  $P \subseteq \mathbb{R}^n$  be a convex polytope. Then there exists a triangulation  $\mathcal{K}$  of  $P$  such that every face of  $P$  is a simplex in  $\mathcal{K}$ .

### 1.4 Dual Polytopes

**Definition 1.4.1 (Polar Dual)**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Define the polar dual of  $P$  to be

$$P^* = \{y \in \mathbb{R}^n \mid \langle x, y \rangle \leq 1 \text{ for all } x \in P\}$$

**Proposition 1.4.2**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Suppose that  $0 \in P$ . Then the following are true.

- $P^*$  is a polytope.

- $P = (P^*)^*$ .
- Let  $F \subseteq P$  be a face. Then  $F^*$  is a face of  $P^*$  of dimension  $\dim(P) - 1 - \dim(F)$ .
- There is a inclusion reversing bijection  $\mathcal{F}(P) \xrightarrow{1:1} \mathcal{F}(P)^{\text{op}}$  given by  $F \mapsto F^*$ .

**Definition 1.4.3 (Dual Graph)**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Define the dual graph  $G^*(P)$  as follows.

- $V(G^*(P)) = \{F \subseteq P \mid F \text{ is a facet of } P\}$ .
- $E(G^*(P)) = \{(F, G) \mid F \cap G \text{ is a face of dimension } \dim(P) - 2\}$ .

**Lemma 1.4.4**

Let  $P \subseteq \mathbb{R}^n$  be a polytope such that  $0 \in P$ . Then  $G(P^*) = G^*(P)$ .

## 2 Operations on Polytopes

### 2.1 Minkowski Sum

**Definition 2.1.1** (Minkowski Sum)

Let  $P, Q \subseteq \mathbb{R}^n$  be polytopes. Define the Minkowski sum of  $P$  and  $Q$  by

$$P + Q = \{x + y \in \mathbb{R}^n \mid x \in P, y \in Q\}$$

**Proposition 2.1.2**

Let  $P, Q \subseteq \mathbb{R}^n$  be polytopes. Suppose that  $P = \text{Conv}(v_1, \dots, v_p)$  and  $Q = \text{Conv}(w_1, \dots, w_q)$ . Then  $P + Q$  is a polytope and

$$P + Q = \text{Conv}(\{v_i + w_j \mid 1 \leq i \leq p, 1 \leq j \leq q\})$$

### 2.2 The Join of Polytopes

**Definition 2.2.1** (The Join of Two Polytopes)

Let  $P \subseteq \mathbb{R}^n$  and  $Q \subseteq \mathbb{R}^m$  be polytopes. Define the join of  $P$  and  $Q$  as follows. Embed  $P$  into  $\mathbb{R}^{n+m+1}$  by  $x \mapsto (x, 0, \dots, 0)$  and embed  $Q$  into  $\mathbb{R}^{n+m+1}$  by  $y \mapsto (0, \dots, 0, y, 1)$ . Then the join of  $P$  and  $Q$  is

$$P * Q = \{(1 - \lambda)x + \lambda y \mid x \in P, y \in Q, 0 \leq \lambda \leq 1\}$$

**Proposition 2.2.2**

Let  $P$  and  $Q$  be polytopes. Then the following are true.

- $\dim(P * Q) = \dim(P) + \dim(Q) + 1$ .
- $\mathcal{F}(P * Q) = \mathcal{F}(P) * \mathcal{F}(Q)$ .
- $V(P * Q) = V(P) \amalg V(Q)$ .

**Example 2.2.3**

Let  $n, m \in \mathbb{N}^+$ . Then we have

$$\Delta^n * \Delta^m = \Delta^{n+m+1}$$

**Example 2.2.4**

Let  $P$  be a polytope. Then the following are true.

- $P * \Delta^1 \cong \Sigma P$ .
- $P * \Delta^0 \cong SP$ .

### 3 Symmetries of a Polytope

#### 3.1 Congruent Polytopes

**Definition 3.1.1** (Congruent Polytopes)

Let  $P \subseteq \mathbb{R}^n$  and  $Q \subseteq \mathbb{R}^m$  be polytopes. We say that  $P$  and  $Q$  are congruent if there exists a function  $\varphi : P \rightarrow Q$  such that the following are true.

- $\varphi$  is a bijection.
- $\|\varphi(y) - \varphi(x)\| = \|y - x\|$  for all  $x, y \in P$ .

**Proposition 3.1.2**

Let  $P \subseteq \mathbb{R}^n$  and  $Q \subseteq \mathbb{R}^m$  be polytopes. Then  $P$  and  $Q$  are congruent if and only if there exists an isometry  $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^m$  such that  $\varphi(P) = Q$ .

**Definition 3.1.3** (Symmetries of a Polytope)

Let  $P \subseteq \mathbb{R}^n$  be a polytope. A symmetry of  $P$  is a congruence  $\varphi : P \rightarrow P$ . Denote the group of symmetries of  $P$  by

$$\text{Sym}(P) = \{\varphi : P \rightarrow P \mid \varphi \text{ is a congruence}\} \leq \text{Isom}(\mathbb{R}^n)$$

**Lemma 3.1.4**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Let  $F \subseteq P$  be a face of  $P$ . Let  $\varphi : P \rightarrow P$  be a symmetry. Then  $\varphi(F)$  is a face of dimension  $\dim(F)$ .

**Definition 3.1.5** (Vertex Centroid of a Polytope)

**Proposition 3.1.6**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Let  $\tau : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be the map defined by  $\tau(x) = x - \bar{P}$ . Then there is an isomorphism

$$\text{Sym}(P) \cong \tau \text{Sym}(P) \tau^{-1}$$

WLOG we may translate any polytope  $P$  such that  $\bar{P} = 0 \in \mathbb{R}^n$ .

**Proposition 3.1.7**

Let  $P \subseteq \mathbb{R}^n$  be a polytope such that  $\bar{P} = 0$ . Then we have

$$\text{Sym}(P) \leq O(\text{AffSpan}(P)) \cong O(\dim(P), \mathbb{R})$$

#### 3.2 Affine Isomorphisms

**Definition 3.2.1** (Affinely Isomorphic)

Let  $P_1 \subseteq \mathbb{R}^{n_1}$  and  $P_2 \subseteq \mathbb{R}^{n_2}$  be polytopes. We say that  $P_1$  and  $P_2$  are affinely isomorphic if there exists an affine transformation  $f : \mathbb{R}^{n_1} \rightarrow \mathbb{R}^{n_2}$  such that  $f|_{P_1}$  is a bijection.

Note that  $f$  itself need not be an affine isomorphism (i.e. an isomorphism of affine spaces).

**Definition 3.2.2** (Affine Symmetries of a Polytope)

Let  $P \subseteq \mathbb{R}^n$  be a polytope. An affine symmetry of  $P$  is an affine isomorphism  $f : P \rightarrow P$ . Denote the group of affine symmetries of  $P$  by

$$\text{Aff}(P) = \{f : P \rightarrow P \mid f \text{ is an affine isomorphism}\} \leq \text{Aff}(\mathbb{R}^n)$$

**Proposition 3.2.3**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Let  $\tau : \mathbb{R}^n \rightarrow \mathbb{R}^n$  by the map defined by  $\tau(x) = x - \overline{P}$ . Then there is an isomorphism

$$\text{Aff}(P) \cong \tau \text{Aff}(P) \tau^{-1}$$

**Proposition 3.2.4**

Let  $P \subseteq \mathbb{R}^n$  be a polytope such that  $\overline{P} = 0$ . Then we have

$$\text{Aff}(P) \leq O(\text{AffSpan}(P)) \cong \text{GL}(\dim(P), \mathbb{R})$$

**Proposition 3.2.5**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Then we have

$$\text{Sym}(P) \leq \text{Aff}(P) \leq \text{Aut}(\mathcal{F}(P))$$

## 4 The Duality between Simple and Simplicial Polytopes

### 4.1 Simplicial Polytopes

#### Definition 4.1.1 (Simplicial Polytopes)

Let  $P \subseteq \mathbb{R}^n$  be a polytope. We say that  $P$  is a simplicial polytope if all its facets are simplices.

#### Lemma 4.1.2

Let  $P \subseteq \mathbb{R}^n$  be a simplicial polytope. Then  $\partial P$  is a simplicial complex.

#### Lemma 4.1.3

Let  $P \subseteq \mathbb{R}^n$  be a simplicial polytope. Then  $\partial P$  is a triangulation of  $S^{\dim(P)-1}$ .

#### Proposition 4.1.4 (The Dehn-Sommerville Equation)

Let  $P \subseteq \mathbb{R}^n$  be a simplicial polytope of dimension  $d$ . Then the following are true:

$$\sum_{j=k}^{d-1} (-1)^j \binom{j+1}{k+1} f_j = (-1)^{d-1} f_k$$

for  $-1 \leq k \leq d-2$ .

In particular,  $\partial P$  is a simplicial complex. The notion of face vector coincides, and they enjoy both the Dehn-Sommerville equation and its relation with the  $h$ -vector.

#### Example 4.1.5

Substituting  $k = -1$  in the Dehn-Sommerville equation recovers Euler's identity.

#### Example 4.1.6

Let  $P$  be a simplicial 3-polytope. Then the Dehn-Sommerville equations give

$$2 = f_0 - f_1 + f_2 \quad \text{and} \quad 2f_1 = 3f_2$$

Substituting gives  $f_1 = 3f_0 - 6$ . Hence the  $f$ -vector of  $P$  is given by

$$f(P) = (f_0, 3f_0 - 6, 2f_0 - 4)$$

#### Example 4.1.7

Let  $P$  be a simplicial 4-polytope. Then the Dehn-Sommerville equations give

$$0 = f_0 - f_1 + f_2 - f_3 \quad \text{and} \quad 2f_0 + 3f_2 = 2f_1 + 4f_3 \quad \text{and} \quad 2f_3 = f_2$$

Substituting gives  $f_1 = 3f_0 - 6$ . Hence the  $f$ -vector of  $P$  is given by

$$f(P) = (f_0, f_1, 2(f_1 - f_0), f_1 - f_0)$$

### 4.2 Simple Polytopes

#### Definition 4.2.1 (Simple Polytopes)

Let  $P \subseteq \mathbb{R}^n$  be a polytope. We say that  $P$  is a simple polytope if for all vertices  $v \in P$ , there exists  $n$  edges  $e_1, \dots, e_n \subseteq P$  such that  $v \in \partial e_i$  for all  $i$ .

#### Lemma 4.2.2

Let  $P_1 \subseteq \mathbb{R}^n$  and  $P_2 \subseteq \mathbb{R}^m$  be simple polytopes. Then the Cartesian product  $P_1 \times P_2$  is a simple polytope.

### 4.3 The Duality Theorem

**Proposition 4.3.1**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Then  $P$  is simple if and only  $P^*$  is simplicial.

Duality of  $d$ -dimensional faces with  $(n - d)$ -dimensional faces.

**Proposition 4.3.2**

Let  $P \subseteq \mathbb{R}^n$  be a polytope. If  $P$  is both simple and simplicial, then  $P$  belongs to one of the following types of polytopes.

- $n = 2$  (and so  $P$  is a polygon).
- $P$  is a simplex.

### 4.4 The Nerve Complex

**Definition 4.4.1** (The Nerve Complex)

Let  $P \subseteq \mathbb{R}^n$  be a simple polytope. Define the nerve complex of  $P$  to be the simplicial complex

$$\mathcal{K}_P = \partial(P^*)$$

**Lemma 4.4.2**

Let  $P_1 \subseteq \mathbb{R}^n$  and  $P_2 \subseteq \mathbb{R}^m$  be simple polytopes. Then we have

$$\mathcal{K}_{P_1 \times P_2} \cong \mathcal{K}_{P_1} * \mathcal{K}_{P_2}$$

## 5 Extremal Polytopes

### 5.1 Cyclic Polytopes

#### Definition 5.1.1 (Cyclic Polytopes)

Let  $\gamma : \mathbb{R} \rightarrow \mathbb{R}^d$  be the function  $\gamma(t) = (t, \dots, t^d)$ . Let  $t_1, \dots, t_n \in \mathbb{R}$  such that  $t_i < t_{i+1}$  for  $1 \leq i \leq n-1$ . Define the cyclic polytope of dimension  $d$  with  $n$  vertices by

$$C(n, d) = \text{Conv}(\{\gamma(t_1), \dots, \gamma(t_n)\})$$

Note: different choices of  $t_1, \dots, t_n$  lead to combinatorially equivalent cyclic polytopes.

#### Lemma 5.1.2

Let  $n, d \in \mathbb{N}^+$ . Then  $C(n, d)$  is a simplicial polytope.

#### Theorem 5.1.3 (Gale Evenness Condition)

Let  $n, d \in \mathbb{N}^+$ . Let  $t_1, \dots, t_n \in \mathbb{R}$  such that

$$C(n, d) = \text{Conv}(\{\gamma(t_1), \dots, \gamma(t_n)\})$$

Then  $\text{Conv}(\{t_{i_1}, \dots, t_{i_d}\})$  is a facet of  $C(n, d)$  if and only if  $i_k - i_j$  is an odd number for  $1 \leq j, k \leq d$ .

#### Proposition 5.1.4

Let  $n, d \in \mathbb{N}^+$ . Then we have

$$f_{\lfloor d/2 \rfloor - 1}(C(n, d)) = \binom{n - \lceil d/2 \rceil}{\lfloor d/2 \rfloor} + \binom{n - \lfloor d/2 \rfloor - 1}{\lceil d/2 \rceil - 1}$$

#### Lemma 5.1.5

Let  $n, d \in \mathbb{N}^+$ . Then we have

$$f_i(C(n, d)) = \binom{n}{i+1}$$

for  $0 \leq i \leq \lfloor \frac{n}{2} \rfloor - 1$ .

#### Corollary 5.1.6

Let  $n, d \in \mathbb{N}^+$ . Then we have

$$f_i(C(n, d)) = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{k}{d-1-i} \binom{n-d+k-1}{k} + \sum_{j=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{d-j}{d-1-i} \binom{n-d+j-1}{j}$$

for  $0 \leq i \leq d$ , where we set  $\binom{p}{q} = 0$  if  $p < q$  or  $q < 0$ .

#### Theorem 5.1.7 (Upper Bound Theorem)

Let  $P$  be a polytope of dimension  $d$ . Suppose that  $f_0(P) = n$ . Then we have

$$f_k(P) \leq f_k(C(n, d))$$

for all  $k$ .

## 5.2 Stacked Polytopes

### Definition 5.2.1 (Vertex Pulling)

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Let  $F \subseteq P$  be a face of  $P$ . Let  $v \in \mathbb{R}^n \setminus P$  such that the following are true.

- Suppose that  $H$  is a supporting hyperplane of  $F$ . Then  $v$  and  $P$  are on different sides in  $\mathbb{R}^n \setminus H$ .
- Suppose that  $G \subseteq P$  is a facet such that  $G \neq F$ , and  $H$  is a supporting hyperplane of  $G$ . Then  $v$  and  $P$  are on the same side  $\mathbb{R}^n \setminus H$ .

Define the pulling of  $P$  by  $v$  along  $F$  to be the polytope

$$P' = \text{Conv}(P \cup \{v\})$$

### Definition 5.2.2 (Stacked Polytopes)

Let  $P \subseteq \mathbb{R}^n$  be a polytope. We say that  $P$  is stacked if there exists a sequence of polytopes  $\Delta^{\dim(P)} = S_0, \dots, S_k = P$  such that  $S_{i+1}$  is obtained from  $S_i$  by vertex pulling for all  $0 \leq i \leq k-1$ .

### Proposition 5.2.3

Let  $P \subseteq \mathbb{R}^n$  be a stacked polytope. Then the following are true.

- $P$  is obtained from  $\Delta^{\dim(P)}$  by  $|V(P)| - \dim(P) - 1$  vertex pullings.
- $P$  is simplicial.
- The  $f$ -vector of  $P$  is given by

$$f_k(P) = \binom{\dim(P)}{k} V(P) - \binom{\dim(P) + 1}{k + 1} k$$

for  $1 \leq k \leq \dim(P) - 1$ .

### Proposition 5.2.4

Let  $P \subseteq \mathbb{R}^n$  be a simplicial polytope such that  $\dim(P) \geq 2$ . Then the following are equivalent.

- $P$  is a stacked polytope.
- $G^*(P)$  is a tree.
- If  $F \in \text{MF}(\partial P)$  is a missing face, then  $\dim(F) = \left\lfloor \frac{\dim(P)}{2} \right\rfloor$ .

When  $\dim(P) = 3$ , then  $P$  is a stacked polytope if and only if  $\partial P$  is a flag complex.

### Theorem 5.2.5 (Lower Bound Theorem)

Let  $P$  be a simplicial polytope such that  $\dim(P) \geq 3$ . Suppose that  $f_0(P) = n$ . Let  $Q$  be a stacked polytope with  $n$  vertices. Then we have

$$f_k(P) \geq f_k(Q)$$

for all  $k$ .

## 5.3 Neighbourly Polytopes

Let  $\mathcal{K}$  be a simplicial complex. Recall that  $\mathcal{K}$  is  $k$ -neighbourly if for any  $I \subseteq V(\mathcal{K})$  such that  $|I| \leq k$ , we have that  $I \in \mathcal{K}$ . This definition also applies to polytopes.

## 6 Classical Families of Polytopes

### 6.1 Cross Polytopes

### 6.2 Regular Polytopes

#### Definition 6.2.1 (Flags of a Polytope)

Let  $P \subseteq \mathbb{R}^n$  be a polytope. A flag of  $P$  is a maximal chain in  $\mathcal{F}(P)$ . Denote the set of flags of  $P$  by

$$\text{Flag}(P) = \{\Phi \subseteq \mathcal{F}(P) \mid \Phi \text{ is a flag of } P\}$$

Since symmetries of a polytope sends faces to faces of the same dimension, we obtain a group action of  $\text{Sym}(P)$  on  $\text{Flag}(P)$ .

#### Definition 6.2.2 (Regular Polytopes)

Let  $P \subseteq \mathbb{R}^n$  be a polytope. We say  $P$  is a regular polytope if  $\text{Sym}(P)$  acts transitively on  $\text{Flag}(P)$ .

**Lemma 6.2.3** | Let  $P \subseteq \mathbb{R}^n$  be a regular polytope. Then  $P^*$  is regular.

#### Example 6.2.4 (Regular n-Polygons)

Let  $n \geq 3$ . Define the regular  $n$ -gon to be

$$P_n = \text{Conv} \left( \left\{ \left( \cos \left( \frac{2\pi k}{n} \right), \sin \left( \frac{2\pi k}{n} \right) \right) \mid 0 \leq k \leq n-1 \right\} \right)$$

Then there is an isomorphism

$$\text{Sym}(P_n) \cong D_{2n}$$

#### Proposition 6.2.5

Let  $P$  be a regular polytope of dimension 1. Then  $P$  is congruent to  $[-1, 1]$ .

#### Proposition 6.2.6

Let  $P$  be a regular polytope of dimension 2. Then  $P$  is congruent to  $P_n$  for some  $n \geq 3$ .

#### Proposition 6.2.7

Let  $P$  be a regular polytope of dimension 3. Then  $P$  is congruent to exactly one of the following.

- The tetrahedron.
- The cube.
- The octahedron.
- The dodecahedron.
- The icosahedron.

## 7 Piecewise Linear Structure

### 7.1 Relation between Polyhedra and Simplicial Complexes

#### Definition 7.1.1 (Locally Finite Collections)

Let  $X$  be a space. Let  $P \subseteq \mathcal{P}(X)$  be a collection of subsets. We say that  $P$  is locally finite if for each  $p \in X$ , there exists a neighbourhood  $U \subseteq X$  of  $p$  such that the set

$$\{V \in P \mid U \cap V \neq \emptyset\}$$

is finite.

#### Proposition 7.1.2

Let  $P$  be a polyhedra. Then  $P$  is a locally finite union of simplexes.

Thus if  $P$  is a polytope, then  $P$  has the underlying structure of a simplicial complex.

#### Proposition 7.1.3

Let  $P \subseteq \mathbb{R}^n$  be a polytope. Then there is a PL homeomorphism

$$P \cong_{\text{PL}} D^{\dim(P)}$$

### 7.2 Piecewise Linear Maps between Polyhedra

#### Proposition 7.2.1

Let  $P, Q$  be polyhedra. Let  $f : P \rightarrow Q$  be a function. Then the following are equivalent.

- $f$  is piecewise linear.
- For each  $p \in P$ , there exists  $L \subseteq P$  such that  $f(\lambda p + \mu x) = \lambda f(p) + \mu f(x)$  for all  $\lambda, \mu \geq 0$  and  $\lambda + \mu = 1$  and  $x \in L$ .
- $\Gamma(f) = \{(x, f(x)) \mid x \in P\}$  is a polyhedron.
- There exists a locally finite decomposition into polyhedra  $P = \bigcup_{\alpha \in I} P_\alpha$  such that for each  $\alpha \in I$ , the restriction  $f|_{P_\alpha}$  is affine linear.

#### Lemma 7.2.2

Let  $P, Q$  be polyhedra. Let  $f : P \rightarrow Q$  be a piecewise linear function. If  $f$  is a homeomorphism, then  $f$  is a piecewise linear homeomorphism.

### 7.3 Polytopal Spheres

#### Definition 7.3.1 (Polytopal Spheres)

Let  $S$  be a simplicial complex. We say that  $S$  is a polytopal sphere if  $S$  is the boundary of a polytope.

Together with results from Simplicial Topology, we have a series of inclusions

$$\text{Polytopal Spheres} \subseteq \text{Combinatorial Spheres} \subseteq \text{Simplicial Spheres}$$

When  $\dim = 2$ , all inclusions are equalities. When  $\dim = 3$ , the first inclusion is strict and the second inclusion is an equality. At  $\dim = 4$ , the second inclusion is an open problem, at  $\dim \geq 5$ , the inclusion is strict. If the number of vertices is  $\leq \dim + 4$ , then all inclusions are equalities.

## 8 Cones and Fans

### 8.1 Cones and its Properties

#### Definition 8.1.1 (Cones)

Let  $C \subseteq \mathbb{R}^n$ . We say that  $C$  is a cone if  $x \in C$  implies that  $sx \in C$  for all  $s \geq 0$ .

#### Definition 8.1.2 (Polyhedral Cones)

A cone in  $\mathbb{R}^n$  is a subset  $C \subseteq \mathbb{R}^n$  such that if  $c_1, \dots, c_n \in C$ , then  $\sum_{i=1}^n \lambda_i c_i \in C$  for all  $\lambda_1, \dots, \lambda_n \geq 0$ .

#### Definition 8.1.3 (Conical Hull)

Let  $S \subseteq \mathbb{R}^n$  be finite. Define the conical hull of  $S$  to be

$$\text{Cone}(S) = \left\{ \sum_{s \in S} \lambda_s s \mid \lambda_s \geq 0 \right\}$$

#### Proposition 8.1.4

Let  $C \subseteq \mathbb{R}^n$  be a subset. Then the following are equivalent.

- $C$  is a polyhedral cone.
- There exists a finite subset  $S \subseteq \mathbb{R}^n$  such that  $C = \text{Cone}(S)$ .
- $C$  is the intersection of finitely many halfspaces containing 0 in their boundary.

Thus the notion of faces, dimensions, supporting hyperplanes etc all apply to polyhedral cones.

#### Definition 8.1.5 (Rays)

Let  $C \subseteq \mathbb{R}^n$  be a cone. We say that  $C$  is a ray with  $\dim(C) = 1$ .

### 8.2 The Dual Cone

#### Definition 8.2.1 (The Dual of a Cone)

Let  $C \subseteq \mathbb{R}^n$  be a cone. Define the dual of  $C$  to be

$$C^\vee = \{x \in \mathbb{R}^n \mid \langle x, c \rangle \geq 0 \text{ for all } c \in C\}$$

#### Definition 8.2.2 (The Dual Face)

Let  $C \subseteq \mathbb{R}^n$  be a cone. Let  $F \subseteq C$  be a face of  $C$ . Define the dual face of  $F$  by

$$F^* = \{x \in C^\vee \mid \langle x, u \rangle = 0 \mid \text{for all } u \in F\}$$

#### Lemma 8.2.3

Let  $C \subseteq \mathbb{R}^n$  be a cone. Let  $F \subseteq C$  be a face of  $C$ . Then the following are true.

- $F^*$  is a face of  $C^\vee$ .
- $F^{**} = F$ .
- $\dim(F^*) = \dim(C) - \dim(F)$ .

#### Proposition 8.2.4

Let  $C \subseteq \mathbb{R}^n$  be a cone. Then there is an inclusion reversing bijection

$$\mathcal{F}(C) \xleftrightarrow{1:1} \mathcal{F}(C^\vee)$$

given by  $F \mapsto F^*$ .

**Definition 8.2.5** (Associated Hyperplane and Halfspace of a Linear Functional)

Let  $\phi \in (\mathbb{R}^n)^*$ . Define the associated hyperplane of  $\phi$  to be

$$H_\phi = \{x \in \mathbb{R}^n \mid \phi(x) = 0\}$$

Define the associated halfspace of  $\phi$  to be

$$H_\phi^+ = \{x \in \mathbb{R}^n \mid \phi(x) \geq 0\}$$

**Lemma 8.2.6**

Let  $C \subseteq \mathbb{R}^n$  be a cone. Let  $H$  be a hyperplane in  $\mathbb{R}^n$ . Then  $H$  is a supporting hyperplane of  $C$  if and only if there exists  $\phi \in C^\vee$  such that  $H = H_\phi$ .

**Lemma 8.2.7**

Let  $C \subseteq \mathbb{R}^n$  be a cone. Then the following are true.

- $C^\vee$  is a cone.
- $(C^\vee)^\vee = C$ .
- $C = \bigcap_{i=1}^n H_{\phi_i}$  for  $\phi_i \in (\mathbb{R}^n)^*$  if and only if  $C^\vee = \text{Cone}(\phi_1, \dots, \phi_n)$ .

**Lemma 8.2.8** (Separation Lemma)

Let  $C_1, C_2 \subseteq \mathbb{R}^n$  be cones such that  $F = C_1 \cap C_2$  is a face of both  $C_1$  and  $C_2$ . Then there exists  $\phi \in C_1^\vee \cap (-C_2)^\vee$  such that

$$F = C_1 \cap H_\phi = C_2 \cap H_\phi$$

### 8.3 Ray Generators

**Definition 8.3.1** (Rational Cones)

Let  $C \subseteq \mathbb{R}^n$  be a cone. We say that  $C$  is rational if there exists a finite set  $S \subseteq \mathbb{Z}^n$  such that  $C = \text{Cone}(S)$ .

**Lemma 8.3.2**

Let  $C \subseteq \mathbb{R}^n$  be a rational cone. Then the following are true.

- If  $F \subseteq C$  is a face of  $C$ , then  $F$  is rational.
- $C^\vee$  is rational.

**Definition 8.3.3** (Strongly Convex Cones)

Let  $C \subseteq \mathbb{R}^n$  be a cone. We say that  $C$  is strongly convex if  $C$  does not contain a line.

**Lemma 8.3.4**

Let  $C \subseteq \mathbb{R}^n$  be a strongly convex cone. Then the following are true.

- If  $F \subseteq C$  is a face of  $C$ , then  $F$  is strongly convex.
- $C^\vee$  is strongly convex.

**Proposition 8.3.5**

Let  $C \subseteq \mathbb{R}^n$  be a cone. Then the following are equivalent.

- $C$  is a strongly convex cone.
- $C \cap (-C) = \{0\}$ .
- $\{0\}$  is a face of  $C$ .
- $\dim(C^\vee) = n$ .

**Definition 8.3.6** (Ray Generators)

Let  $C \subseteq \mathbb{R}^n$  be a strongly convex rational cone. Let  $R_1, \dots, R_m$  be the edges of  $C$ . Define the ray generators of  $C$  to be the unique vectors  $v_1, \dots, v_m \in C$  such that the following are true.

- $v_1, \dots, v_m \in \mathbb{Z}^n$ .
- $v_i \in R_i$  for each  $i$ .
- $C = \text{Cone}(v_1, \dots, v_m)$ .
- For each  $i$ , if  $w_i \in R_i \cap \mathbb{Z}^n$  then  $|v_i| \leq |w_i|$ .

**Definition 8.3.7 (Regular Cones)**

Let  $C \subseteq \mathbb{R}^n$  be a strongly convex rational cone. We say that  $C$  is regular if its ray generators are  $\mathbb{Z}$ -linearly independent.

**Lemma 8.3.8**

Let  $C \subseteq \mathbb{R}^n$  be a regular cone. Then the following are true.

- If  $F \subseteq C$  is a face of  $C$ , then  $F$  is regular.
- $C^\vee$  is regular.

**Definition 8.3.9 (Simplicial Cones)**

Let  $C \subseteq \mathbb{R}^n$  be a strongly convex rational cone. We say that  $C$  is simplicial if its ray generators are  $\mathbb{R}$ -linearly independent.

**Lemma 8.3.10**

Let  $C \subseteq \mathbb{R}^n$  be a simplicial cone. Then the following are true.

- If  $F \subseteq C$  is a face of  $C$ , then  $F$  is simplicial.
- $C^\vee$  is simplicial.

## 8.4 Fans and its Properties

**Definition 8.4.1 (Fans)**

Let  $n \in \mathbb{N}^+$ . A fan in  $\mathbb{R}^n$  is a finite set of cones  $\Sigma$  such that the following are true.

- Each  $C \in \Sigma$  is a strongly convex rational polyhedral cone.
- For each cone  $C \in \Sigma$  and each face  $F$  of  $C$ , we have  $F \in \Sigma$ .
- For cones  $C_1, C_2 \in \Sigma$ , we have  $C_1 \cap C_2$  are faces of  $C_1$  and  $C_2$ .

We write

$$\Sigma(d) = \{C \in \Sigma \mid \dim(C) = d\}$$

for the  $d$ -dimensional cones in the fan.

**Definition 8.4.2 (The Maximal Cones of a Fan)**

Let  $n \in \mathbb{N}^+$ . Let  $\Sigma$  be a fan in  $\mathbb{R}^n$ . Define

$$\Sigma_{\max} = \{\sigma \in \Sigma \mid \sigma \text{ is maximal in } \Sigma \text{ with respect to inclusion}\}$$

**Definition 8.4.3 (Types of Fans)**

Let  $n \in \mathbb{N}^+$ . Let  $\Sigma$  be a fan in  $\mathbb{R}^n$ .

- We say that  $\Sigma$  is rational if all  $\sigma \in \Sigma$  is rational.
- We say that  $\Sigma$  is regular if all  $\sigma \in \Sigma$  is regular.
- We say that  $\Sigma$  is simplicial if all  $\sigma \in \Sigma$  is simplicial.

**Definition 8.4.4 (The Support of a Fan)**

Let  $n \in \mathbb{N}^+$ . Let  $\Sigma$  be a fan in  $\mathbb{R}^n$ . Define the support of  $\Sigma$  to be

$$|\Sigma| = \cup_{C \in \Sigma} C$$

**Definition 8.4.5 (Convex Support)**

Let  $n \in \mathbb{N}^+$ . Let  $\Sigma$  be a fan in  $\mathbb{R}^n$ . We say that  $\Sigma$  has convex support if  $|\Sigma|$  is convex.

**Lemma 8.4.6**

Let  $n \in \mathbb{N}^+$ . Let  $\Sigma$  be a fan in  $\mathbb{R}^n$  with convex support. Then  $|\Sigma|$  is a convex cone.

**Lemma 8.4.7**

Let  $n \in \mathbb{N}^+$ . Let  $\Sigma$  be a fan in  $\mathbb{R}^n$  with convex support and full-dimensional. Then we have

$$|\Sigma| = \text{Cone}(\{u_\rho \mid \rho \in \Sigma(1)\}) = \bigcup_{\sigma \in \Sigma(n)} \sigma$$

**Definition 8.4.8 (Complete Fans)**

Let  $n \in \mathbb{N}^+$ . Let  $\Sigma$  be a fan in  $\mathbb{R}^n$ . We say that  $\Sigma$  is complete if  $|\Sigma| = \mathbb{R}^n$ .

**Lemma 8.4.9**

Let  $n \in \mathbb{N}^+$ . Let  $\Sigma$  be a fan in  $\mathbb{R}^n$ . If  $\Sigma$  is complete, then  $\Sigma_{\max} = \Sigma(n)$ .

## 8.5 The Normal Fan of a Polytope

**Definition 8.5.1 (The Normal Fan of a Polytope)**

Let  $P$  be a convex polytope. Define the normal fan of  $P$  to be

$$\Sigma_P = \{\sigma_Q \mid Q \text{ is a face of } P\}$$

where

$$\sigma_Q = \{x \in \mathbb{R}^n \mid \langle q, x \rangle \leq \langle p, x \rangle \text{ for all } p \in P \text{ and } q \in Q\}$$

**Lemma 8.5.2**

Let  $P$  be a convex polytope. Then  $\Sigma_P$  is a complete fan.

**Lemma 8.5.3**

Let  $P \subseteq \mathbb{R}^n$  be a convex polytope. Then the following are true.

- For any  $m \in \mathbb{R}^n$ , we have  $\Sigma_P = \Sigma_{m+P}$ .
- For any  $k > 0$ , we have  $\Sigma_P = \Sigma_k P$ .

**Definition 8.5.4 (Delzant Polytopes)**

Let  $P$  be a convex polytope. We say that  $P$  is Delzant if  $\Sigma_P$  is regular.

## 9 Lattice Polytopes

### 9.1 Basic Definitions

**Definition 9.1.1** (Lattice Polytopes)

Let  $P \subseteq \mathbb{R}^n$  be a convex polytope. We say that  $P$  is a lattice polytope if there exists a lattice  $N \subseteq \mathbb{R}^n$  such that  $V(P) \subseteq N$ .

### 9.2 The Fan of a Full-Dimensional Lattice Polytope

Using a coordinate transformation, we may always assume that  $V(P) \subseteq \mathbb{Z}^n$  for a lattice polytope  $P$ . For a lattice polytope, for each facet  $F = H_{x_i, b}^+ \cap P$  of  $P$ , the ray generated by  $x_i$  is rational and we can always find a ray generator. Thus  $P$  admits a canonical presentation  $P = \bigcap_{i=1}^k H_{u_F, b}^+$  by these generators  $u_F$ .

Let  $\Sigma_P = \{\sigma_Q \mid Q \text{ is a face of } P\}$  be the normal fan of  $P$ . If  $P$  is full dimensional, we have

$$\sigma_Q = \text{Cone}(\{u_F \mid Q \subseteq F\})$$

**Proposition 9.2.1**

Let  $P$  be a full-dimensional lattice polytope. Then there is an inclusion reversing bijection

$$\mathcal{F}(P) \xleftrightarrow{1:1} \Sigma_P$$

given by  $Q \mapsto \sigma_Q$ .

### 9.3 Normal and Very Ample Polytopes

**Definition 9.3.1** (Normal Polytopes)

Let  $P$  be a lattice polytope. We say that  $P$  is normal if

$$(kP) \cap \mathbb{Z}^n + (lP) \cap \mathbb{Z}^n = ((k+l)P) \cap \mathbb{Z}^n$$

for all  $k, l \in \mathbb{N}$ .

**Proposition 9.3.2**

Let  $P$  be a full-dimensional lattice polytope. If  $\dim(P) \geq 2$ , then  $kP$  is normal for all  $k \geq \dim(P) - 1$ .

**Definition 9.3.3** (Very Ample Polytope)

Let  $P$  be a lattice polytope. We say that  $P$  is very ample if ???

**Proposition 9.3.4**

Let  $P$  be a normal polytope. Then  $P$  is very ample.

**Theorem 9.3.5** (Ewald–Wessels)

Let  $P$  be a full-dimensional lattice polytope. Then  $kP$  is normal for all  $k \geq \dim(P) - 1$ .

### 9.4 Ehrhart Theory

**Definition 9.4.1** (The Ehrhart Polynomial)

Let  $P \subseteq \mathbb{R}^d$  be a full-dimensional lattice polytope. Define the Ehrhart polynomial to be the function  $E_P : \mathbb{N} \rightarrow \mathbb{N}$  given by

$$E_P(m) = |mP \cap \mathbb{Z}^d|$$

**Theorem 9.4.2 (Ehrhart)**

Let  $P \subseteq \mathbb{R}^d$  be a full-dimensional lattice polytope. Then  $E_P \in \mathbb{R}[x]$ .

**Lemma 9.4.3**

Let  $P \subseteq \mathbb{R}^d$  be a full-dimensional lattice polytope. Then the leading term of  $E_P$  is  $\text{Vol}(P)$ .

## 9.5 Reflexive and Gorenstein Polytopes

**Definition 9.5.1 (Reflexive Polytopes)**

Let  $P$  be a lattice polytope. We say that  $P$  is a reflexive polytope if  $P^*$  is a lattice polytope.

## 9.6 Unimodular Triangulations